PCA on graph and multigraph

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2025-09-30

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In this document, we try would try to cluster graph node using a particular PCA on graph, and maybe extend the notion to to cluster node with several graph data.

1 Test on simulation for one graph

In this section, we would like to try the method found in (Saerens et al. 2004) using principal component analysis on a weighted graph.

Let $V = \{1, \dots, d\}$ the set of nodes.

1.1 Method

The procedure use the Laplacian matrix of a weighted graph. Let $\mathcal{G} = (V, E, W)$ an undirected weigthed graph, where W is the weught matrix where $w_{ij} \geq 0$. From this matrix and the set of edges E, we define the adjacency matrix A as:

$$a_{ij} = \begin{cases} w_{ij} & \text{if } (i,j) \in E, \\ 0 & (i,j) \notin E, \end{cases}$$

From this matrix, we are able to define the Laplacian matrix $L = D - A$ where the matrix D is a diagonal matrix where $d_i = \sum_{i=1}^{d} a_{ij}$.

The goal is to use the generalised inverse of the Laplacian matrix L^+ as a new metric called the Euclidian Commute Time Distance (ECTD), and make PCA on this matrix, see (Saerens et al. 2004) for details. It is also linked to the spectral clustering to the graph.

1.2 Adaptation for extremal graphical model

In their example, they used the inverse of the distance between the nodes to set the weights.

The graphical models for extreme produce a graph where an edge does not exist between two nodes if they are independent conditionally to the other nodes. In that sense, we can use weights that measure the strength of extremal dependence between them and mix it with a graph estimation of the vector to being the clustering.

In (Engelke and Volgushev 2022), they see the graph weighted by two notion : the variogram or the extremal coefficient. Tey compute a minimal spanning tree to compute their estimation which are consistent for empirical estimation of these two quantities.

Thus, we will use two possible weights:

$$w_{ij} = \chi_{ij} \qquad \text{or} \qquad w_{ij} = \frac{1}{\Gamma_{ij}}$$

1.3 Application in simulation

For the simulation, we choose a graph of 12 edges with the following structure:

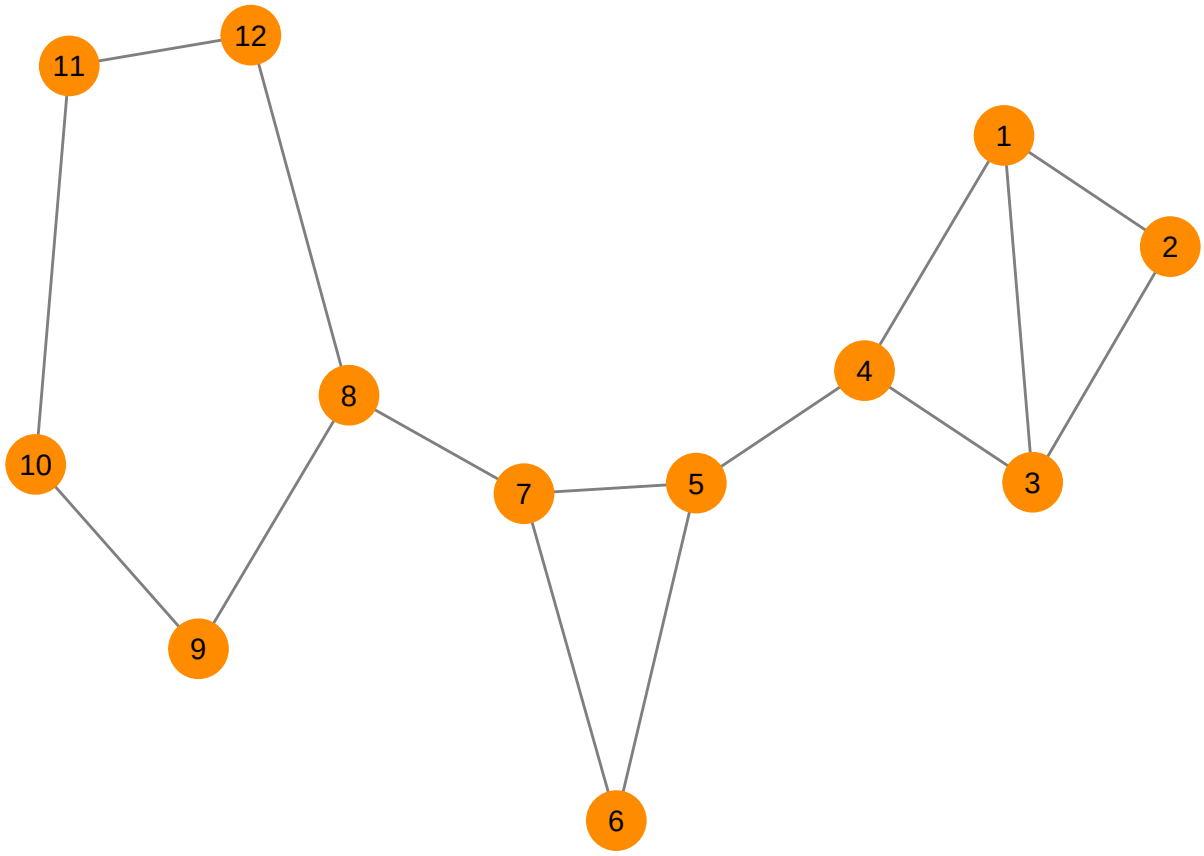


Figure 1: Graphical structure of the simulation.

A first look on this graph suggests us to make the 3 clusters formed by $\{1, 2, 3, 4\}$, $\{5, 6, 7\}$ and $\{8, 9, 10, 11, 12\}$. What clustering will we produce with the procedure ?

1.3.1 Known graph

First, let's see what is going on when we know the graph structure. We assume that the data are Husler-Reiss Pareto distributed, and we take a random variogram with the respect to this graphical structure thanks to `generate_random_graphical_Gamma` function from `graphicalExtremes` package. We get:

And the extremal coefficient matrix:

We can stress that block structure of the matrix seems to gather the node 4 with the second group instead of the first one with only the graphical structure. This observation will be confirmed with the method.

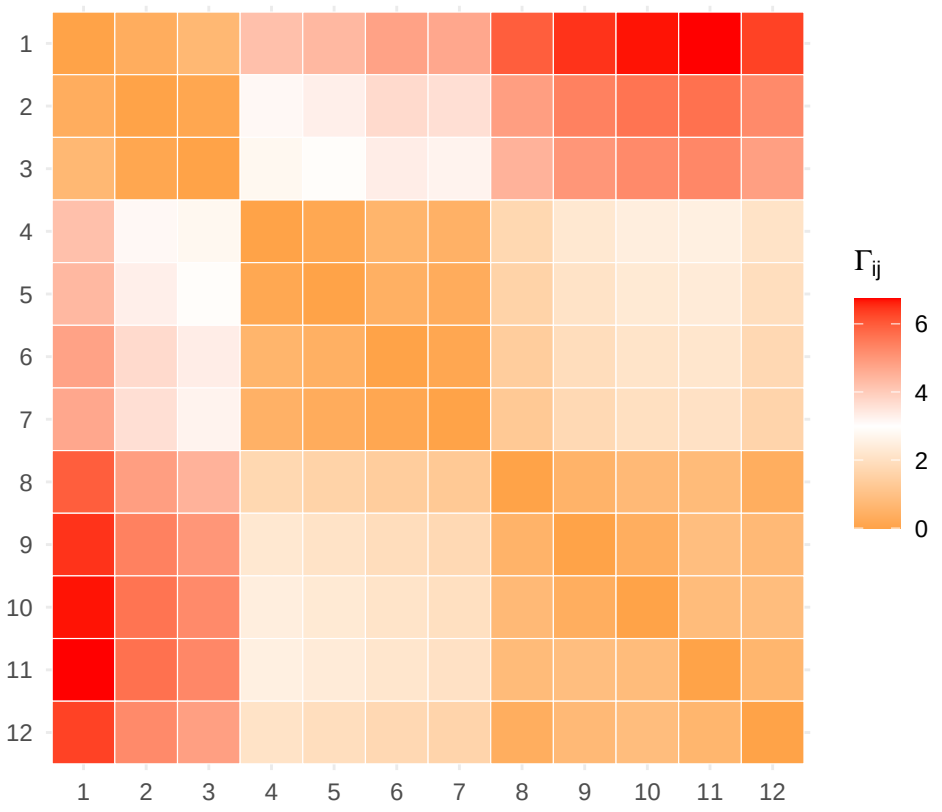


Figure 2: Variogram matrix.

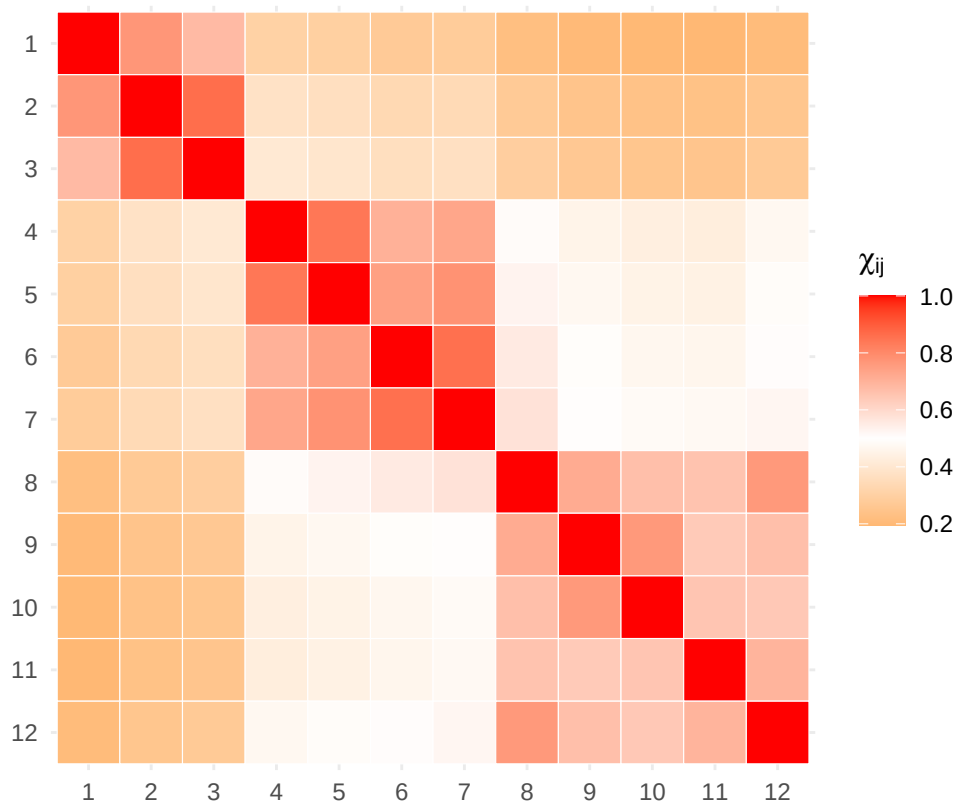


Figure 3: Extremal coefficient matrix.

Indeed, if we compute the PCA on the graph with the corresponding Laplacian matrix we get this for the two firsts principal component for the variogram:

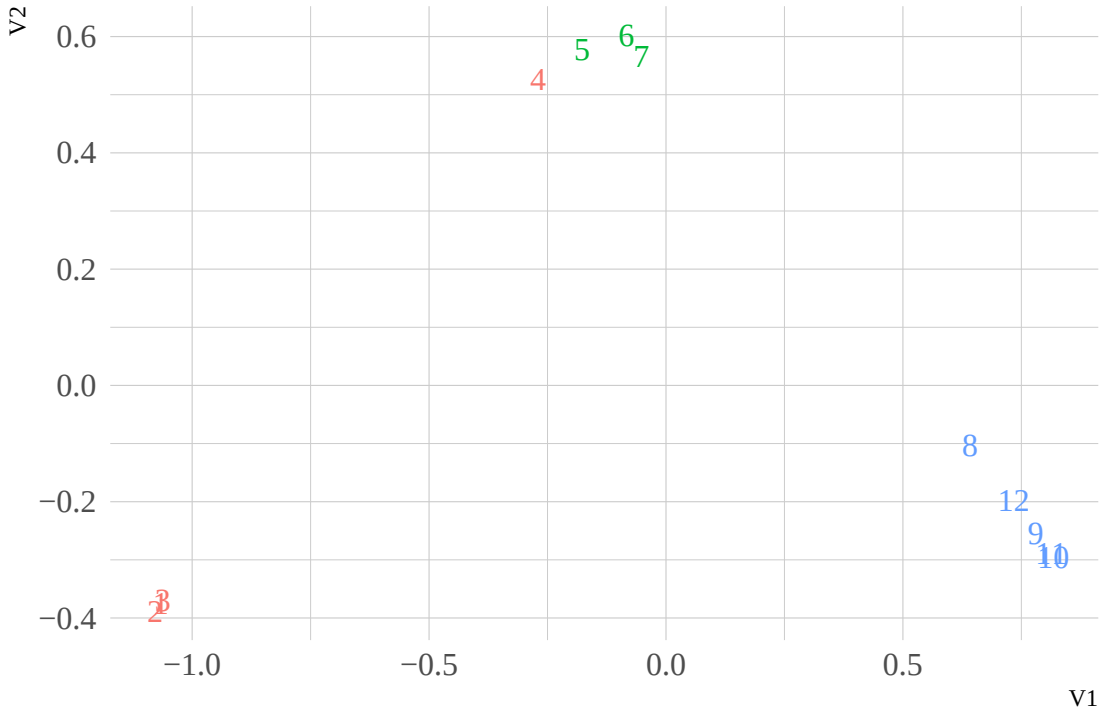


Figure 4: First principal plane with the variogram Laplacian with known graph.

and this for the extremal coefficient:

The colors are made of the initial guess for the clusters. Here, even knowing the true structure, we observe an uncertainty for the node 4, a doubt hightlighted with the extremal coefficient Laplacian.

1.3.2 Unknown graph

With the unknown graph, we need first to estimate the graphical structure. In order to do that, we use `eglearn` function from `graphicalExtremes` introduced in (Engelke, Lalancette, and Volgushev 2024). We get the following graph for a first estimation:

We do not have the right structure here, it can change the results from the previous estimation, and it could erase the encertainty seen for the node 4.

Thus, we get with this graphical structure the following results for the variogram :

and for the extremal coefficient:

As expected, the node 4 is totally associated with the second group. With this first simulation, we can say that the procedure is sensible to the coefficient of the weights, not only on the graphical structure, a conclusion that we could predict.

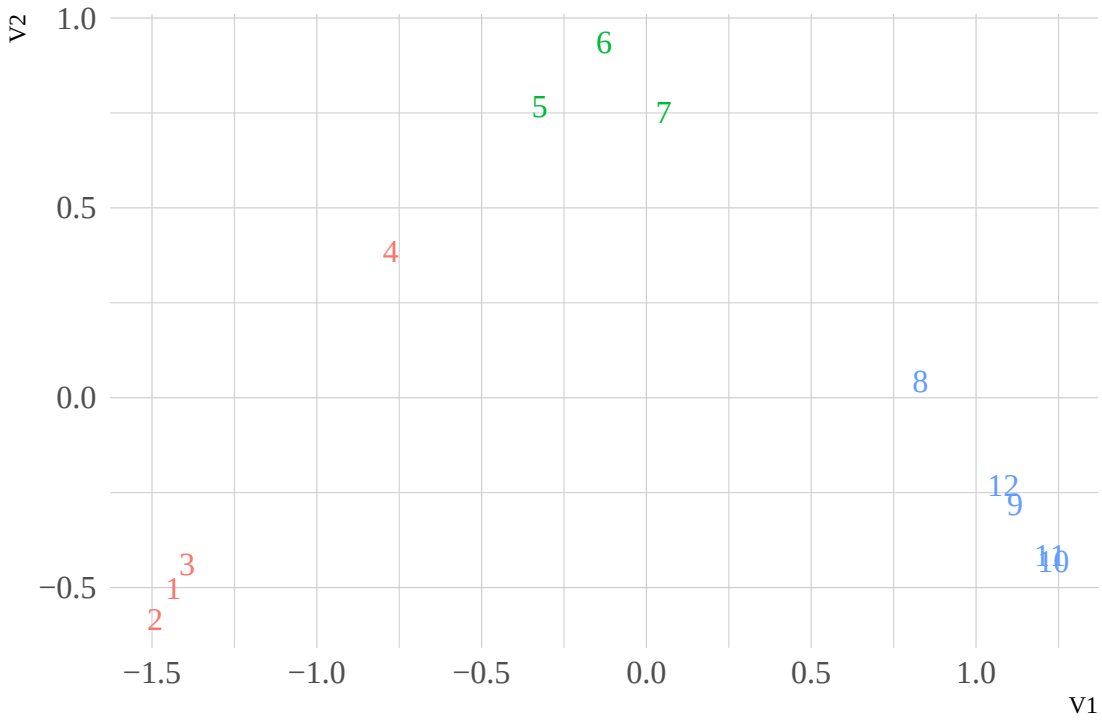


Figure 5: First principal plane with the extremal coefficient Laplacian with known graph.

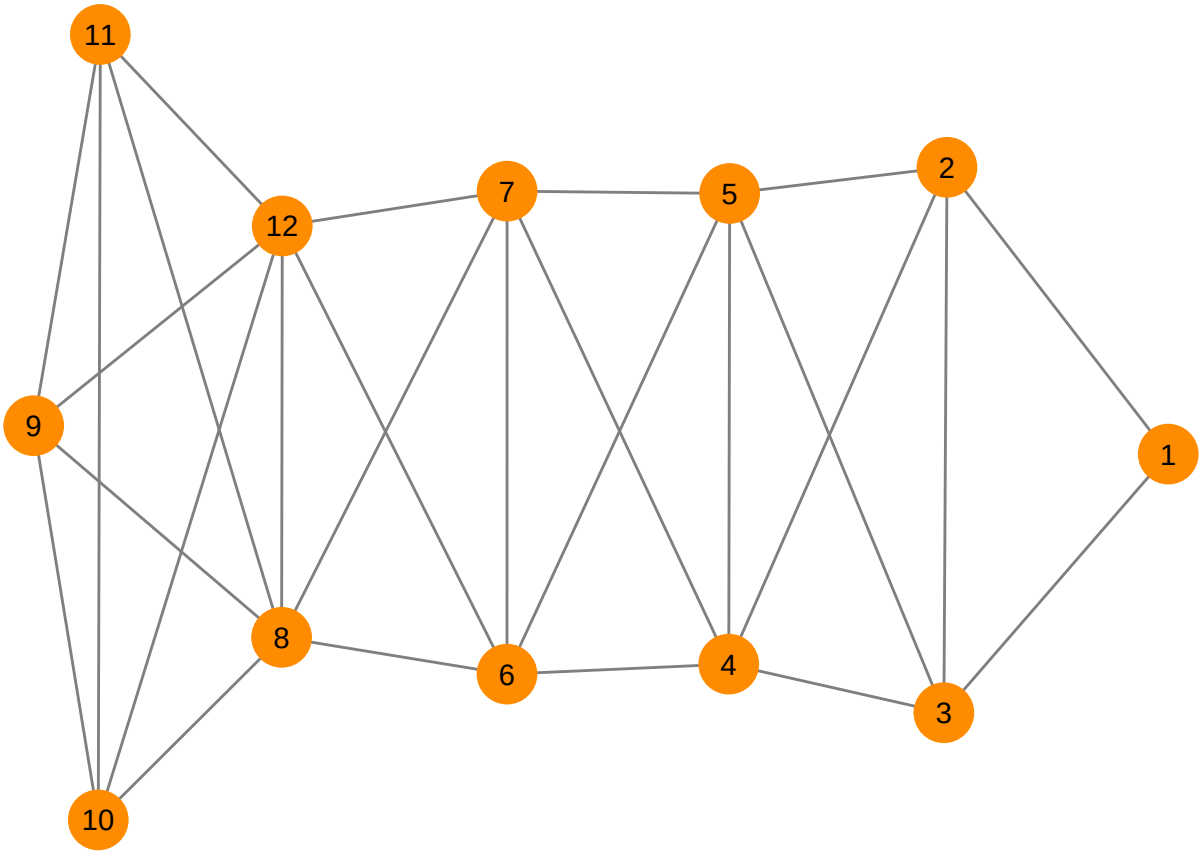


Figure 6: Estimation of the graph.

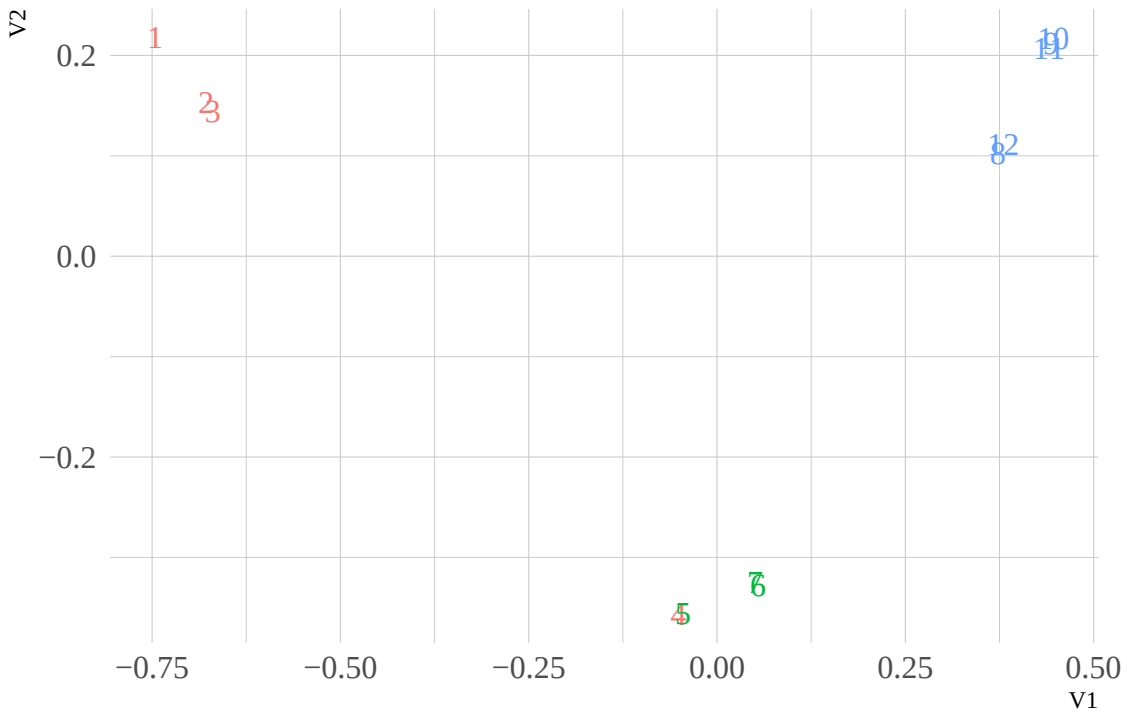


Figure 7: First principal plane with the extremal coefficient Laplacian with unknown graph.

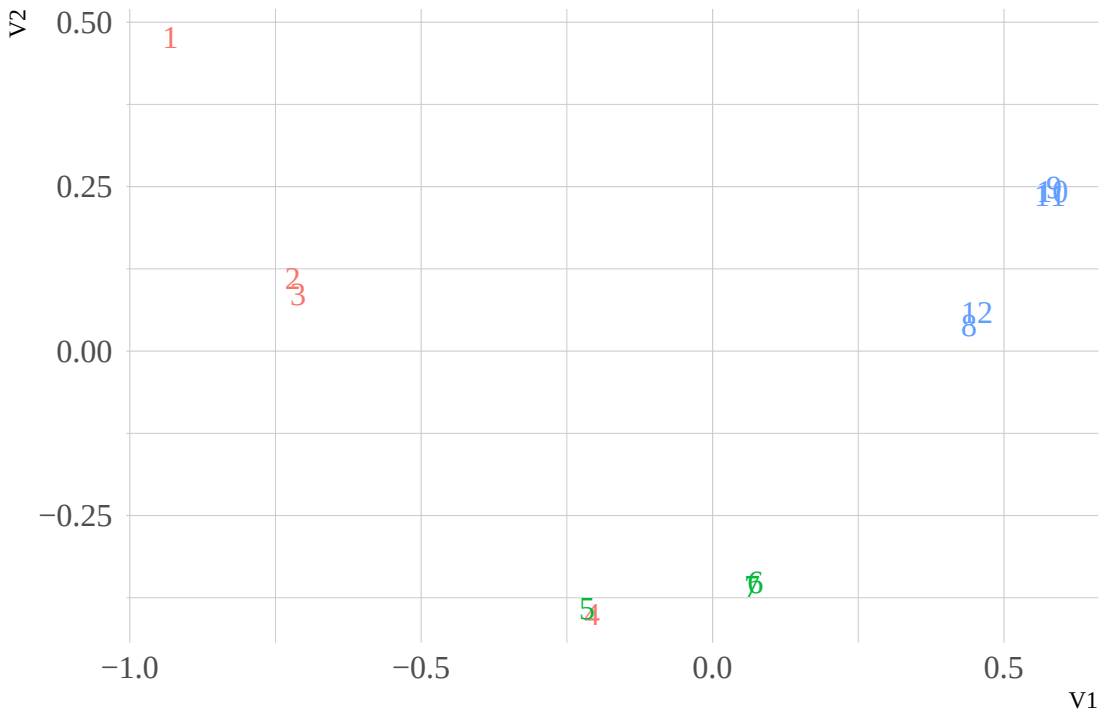
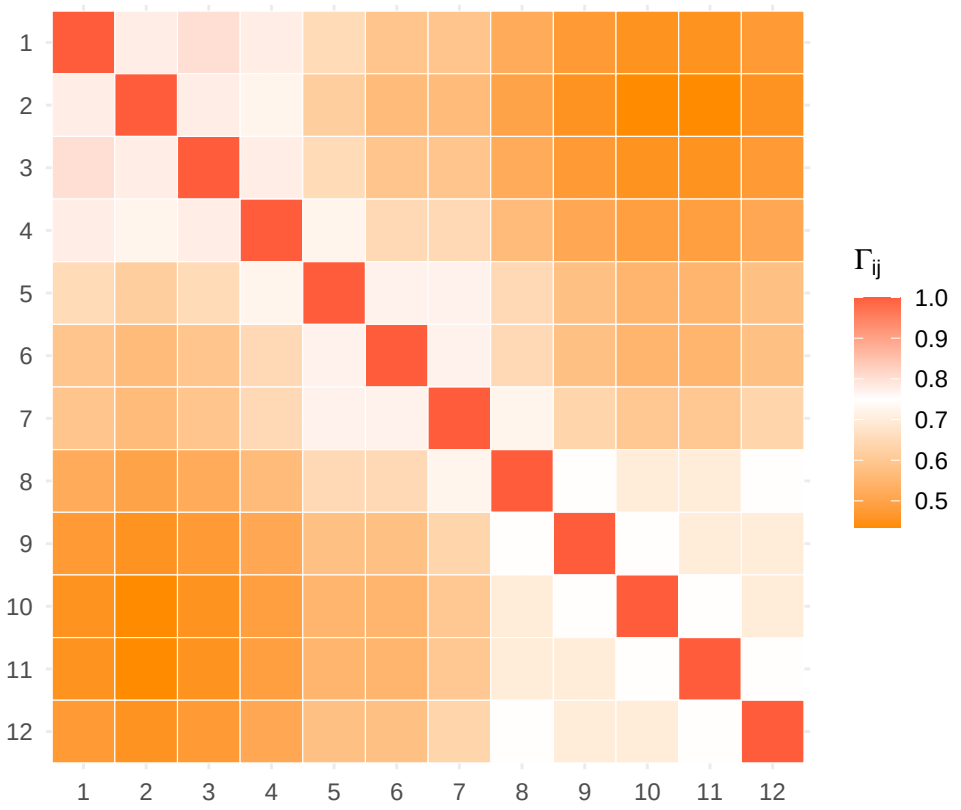


Figure 8: First principal plane with the extremal coefficient Laplacian with unknown graph.

1.3.3 Simulation with all same dependence

We would like to see if the algorithm can separate the nodes even with no clear block matrix structure in the variogram and the extremal coefficient matrix.



The goal is to see if we can see the clustering as an indicator of density of edges between the nodes in the same cluster.

1.4 Application on high dimensionnal graph

Now we would like to challenge the procedure by increasing the dimension. Let's grow the number of variable to $d = 54$ and let's simulate Hüsler-Reiss Pareto distributed data with this graphical structure:

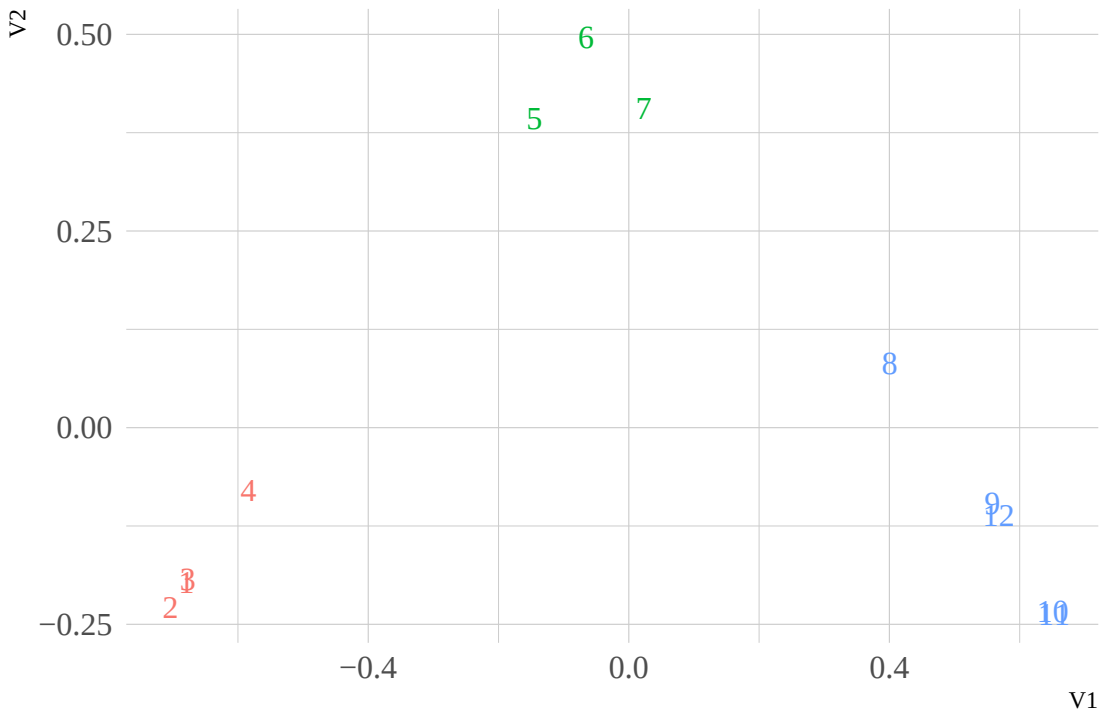
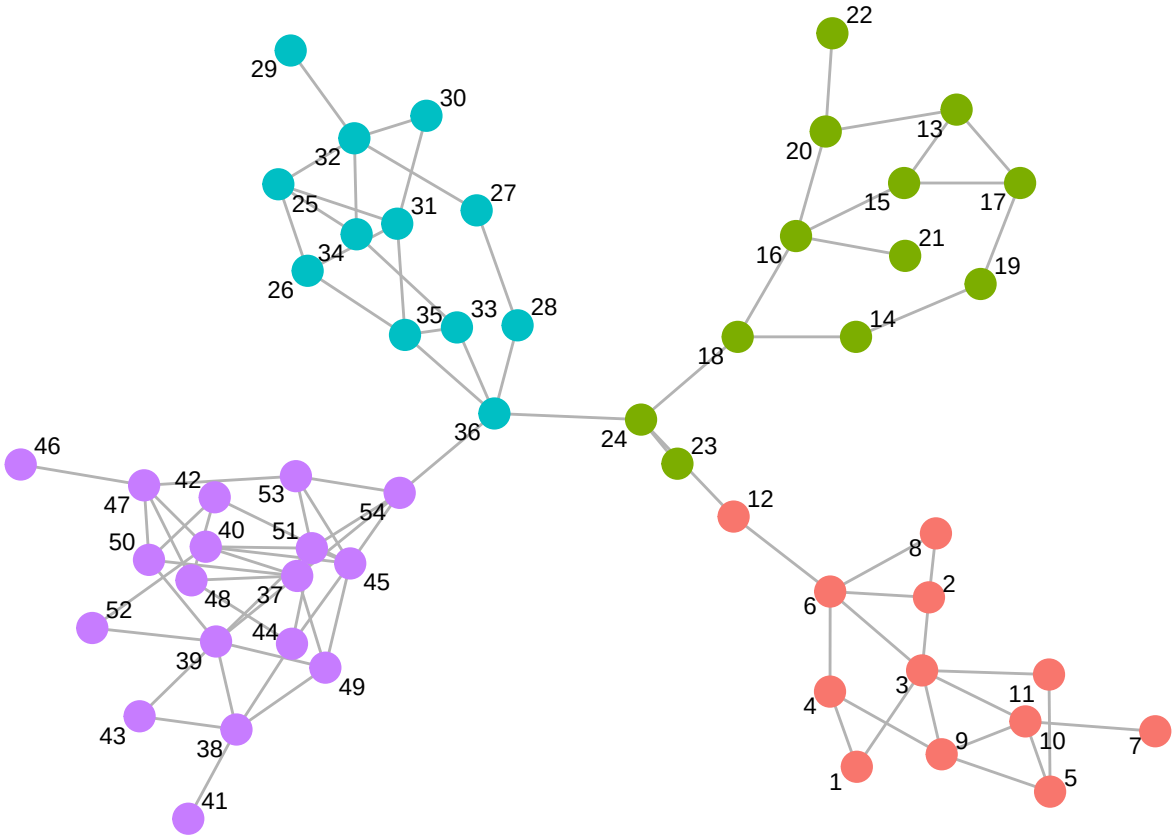


Figure 9: First principal plane with the extremal coefficient Laplacian with known same coefficient graph.



We set the sample size to $n = 200$. After the PCA on the two Laplacian, we end with:

and:

We see stability in the procedure, even with random initialization for the variogram matrix of the model.

The estimation of the graph with `eglearn` gives:

In we get the structure of the blue and purple group. However the red and green group fused totally: we can stress that it exists some ambiguity in the initial structure that can be miss interpretated during estimation. Then, by using this structure in the estimation of the principal component, we get the following first plane:

Again, we see the ambiguity between the two groups after the procedure.

2 Researching method for multigraph

References

Engelke, Sebastian, Michaël Lalancette, and Stanislav Volgushev. 2024. "Learning Extremal Graphical Structures in High Dimensions." arXiv. https://doi.org/10.48550/arXiv.2111.00840.

Engelke, Sebastian, and Stanislav Volgushev. 2022. "Structure Learning for Extremal Tree Models." arXiv. https://doi.org/10.48550/arXiv.2012.06179.

Saerens, Marco, Francois Fouss, Luh Yen, and Pierre Dupont. 2004. "The Principal Components Analysis of a Graph, and Its Relationships to Spectral Clustering." In *Machine Learning: ECML 2004*, edited by Jean-François Boulicaut, Floriana Esposito, Fosca Giannotti, and Dino Pedreschi, 371–83. Berlin, Heidelberg: Springer. https://doi.org/10.1007/978-3-540-30115-8_35.

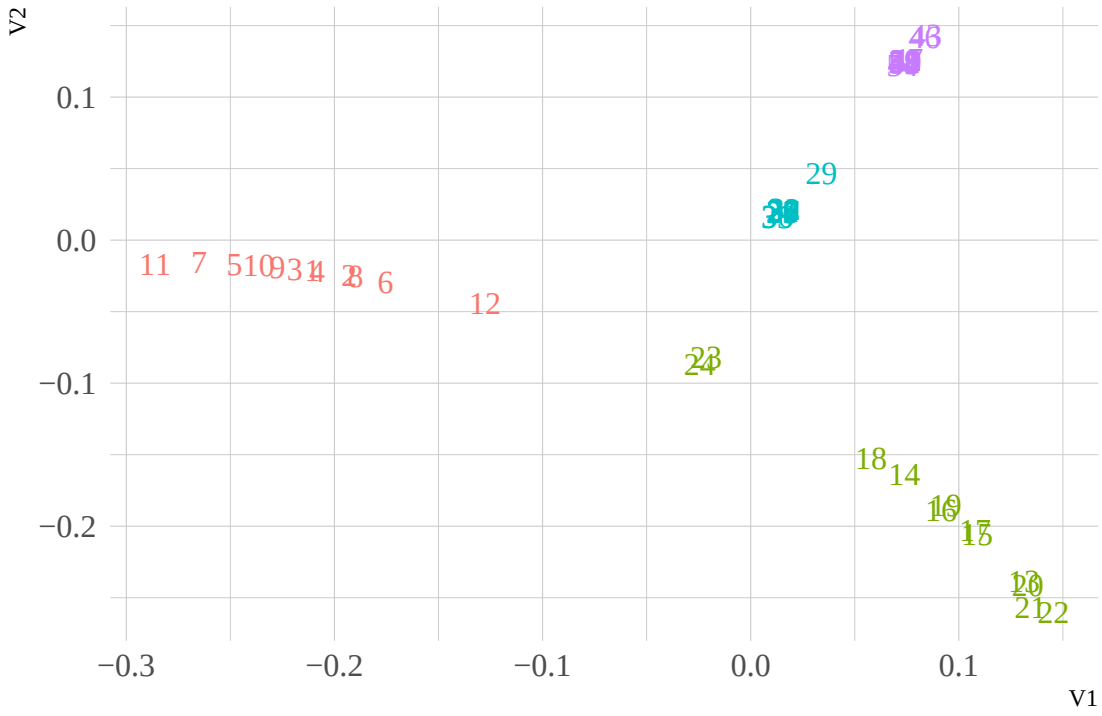


Figure 10: First principal plane with the extremal coefficient Laplacian with known high dimensional graph.

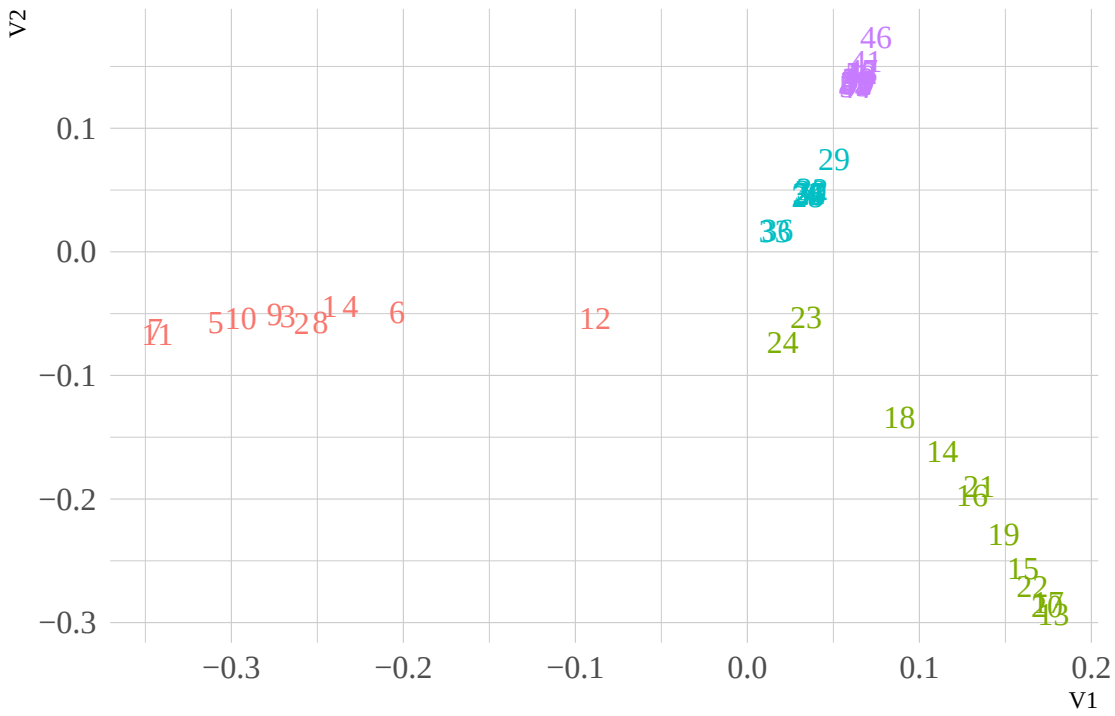


Figure 11: First principal plane with the extremal coefficient Laplacian with known high dimensional graph.

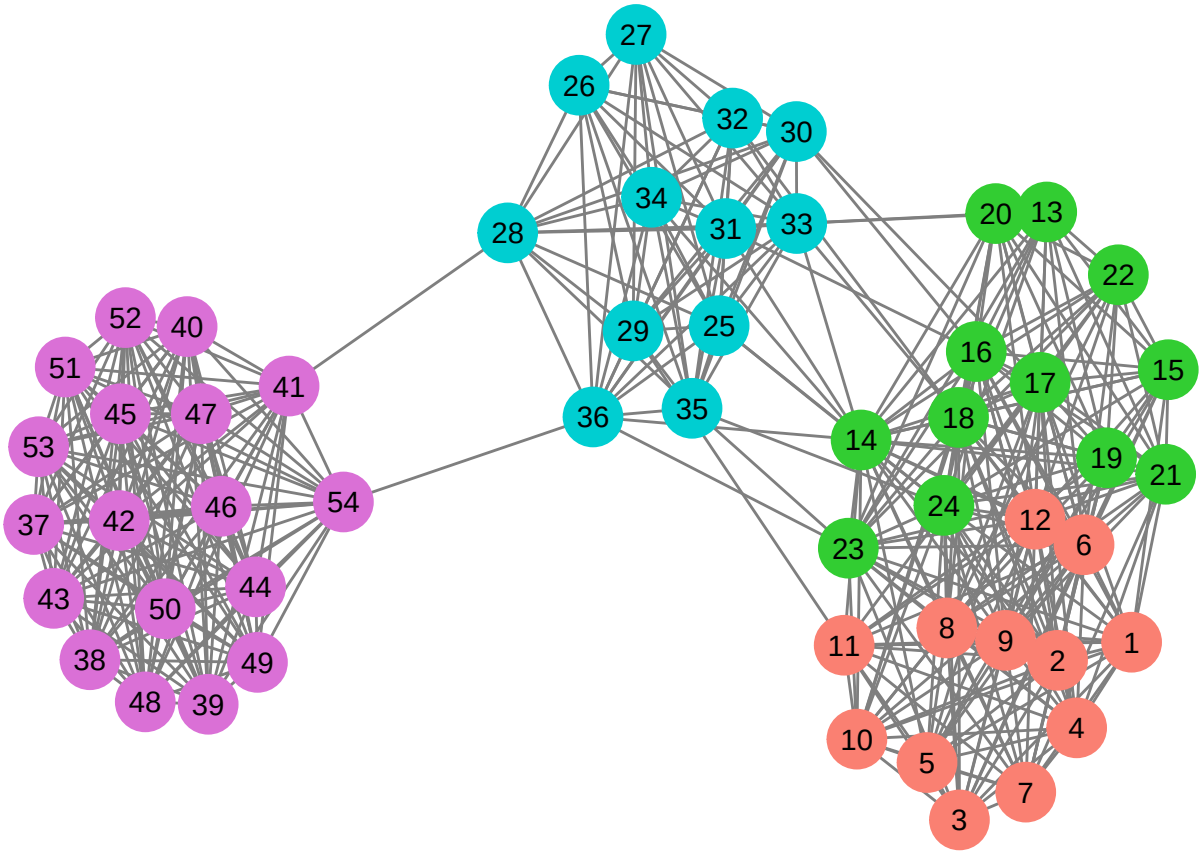


Figure 12: Estimated graph in high dimension.

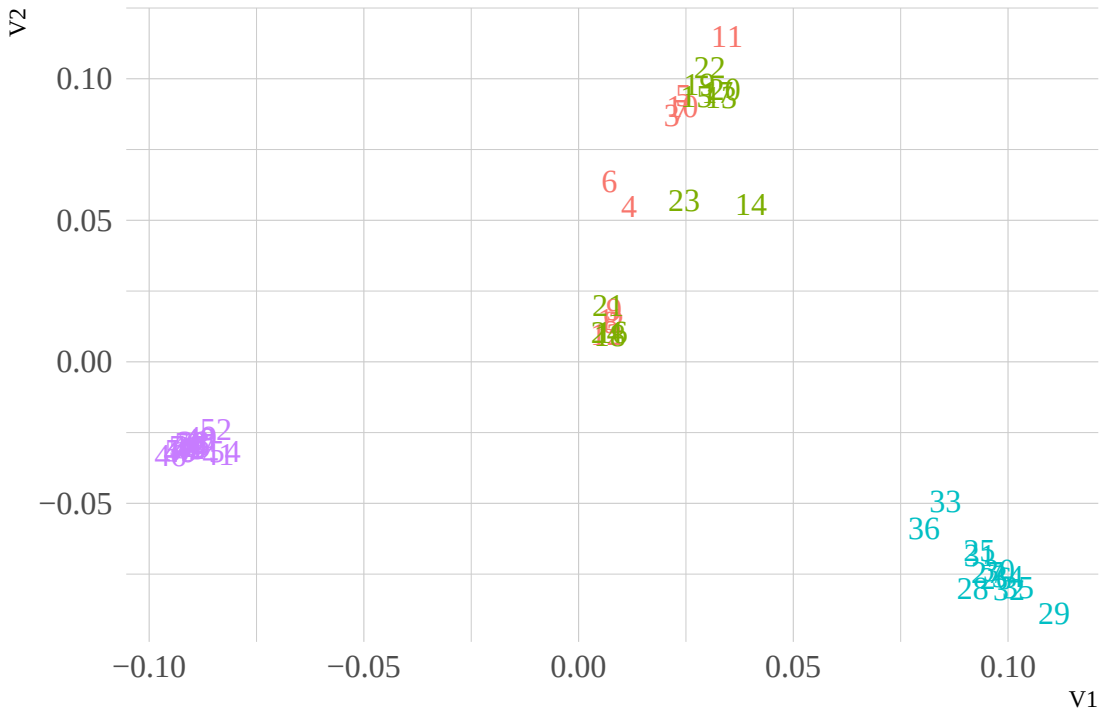


Figure 13: First principal plane with the extremal coefficient Laplacian with unknown high dimensional graph.